The Bilal Morning

Thursday, 15 October 2026 BACKFILLED EDITION 12 stories · 15 sources

2 source articles could not be read today; those stories show headlines and links only.

THREE THINGS TO KNOW

1. SpaceX's first flown Starship upper stage has returned to South Texas for engineering evaluation.
2. The European Space Agency's Hera probe has arrived to investigate the asteroid Dimorphos.
3. Astronaut Frank Rubio has been named as one of the four Artemis III crew members.

First Flown SpaceX Starship Upper Stage Returns Home to Starbase

Nearly three months after launching into space and surviving an ocean splashdown, SpaceX's Ship 40 has returned to South Texas for engineering evaluation.

- Ship 40 is the first Starship upper stage to make it back to its launch site following a flight to space. [space.com]

  > "Ship 40 is the first Starship upper stage to return to its launch site after flying in space"

- The spacecraft survived its splashdown in the Indian Ocean, avoiding the destruction experienced by previous prototypes. [space.com]

  > "Unlike previous Ship splashdowns, though, which have exploded after tipping over into the water, Ship 40 remained largely intact and afloat."

- After being secured in Australia, Ship 40 was ferried back to Texas aboard a semi-submersible heavy-load vessel. [space.com]

  > "Forte arrived at the Port of Brownsville on Oct. 7, wrapping up the longest stretch of Ship 40's return"

- Engineers plan to inspect the vehicle to gather critical data on how its systems held up during atmospheric reentry. [space.com]

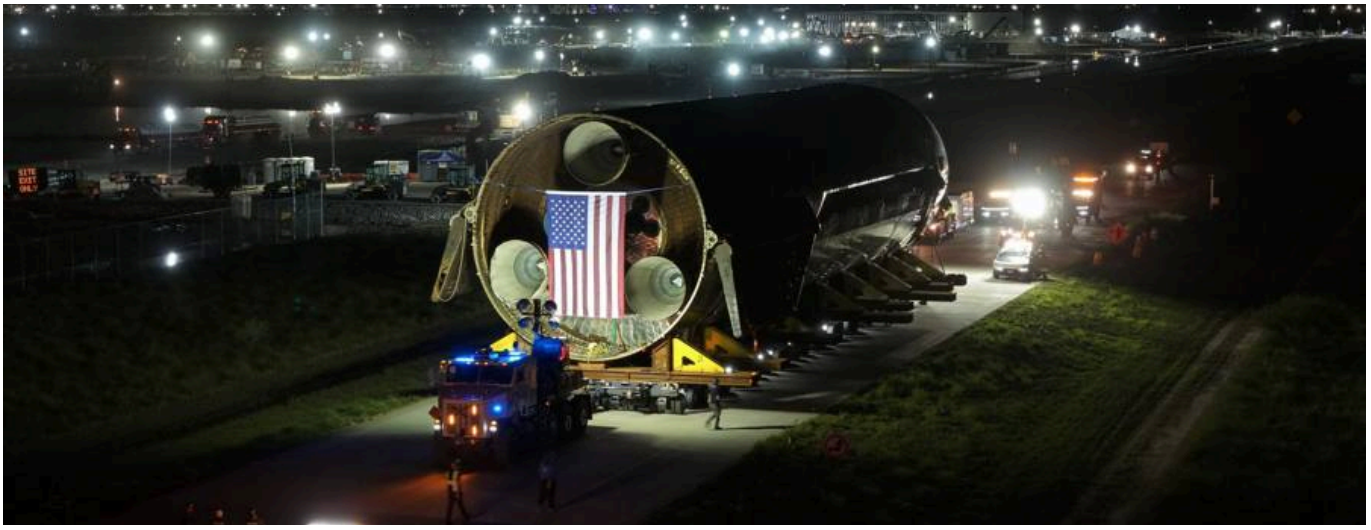
  > "It will provide first-of-its-kind data on a flown Starship heatshield, Raptor engines, avionics, and more,"

In the image: The image captures Ship 40 being transported by road at night on a specialized multi-axle trailer, with a large American flag draped across the open engine section as support vehicles escort the convoy.

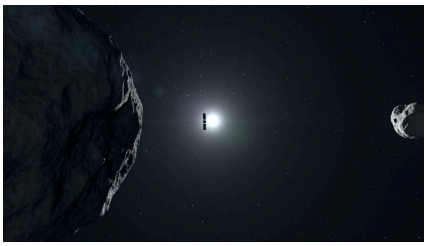
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space.com · theguardian.com · wlky.com · kmbc.com

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SPACE EXPLORATION



European Space Agency's Hera Probe Arrives to Inspect Asteroid Crash

Four years after NASA's DART mission slammed into the asteroid Dimorphos, the Hera spacecraft arrives to investigate the wreckage and study the binary system in detail.

- The DART spacecraft impacted the small asteroid Dimorphos on September 26, 2022, as part of humanity's first planetary defense test. [space.com]

  > "The Double Asteroid Redirection Test (DART) spacecraft slammed into a small space rock called Dimorphos on Sept. 26, 2022"

- The impact successfully shortened Dimorphos's orbit around the larger asteroid Didymos by approximately 33 minutes. [space.com]

  > "definitely succeeded, shortening the little asteroid's orbit around a larger rock called Didymos by about 33 minutes."

- The Hera spacecraft is scheduled to approach the asteroids using three careful braking maneuvers starting mid-October. [space.com]

  > "During three careful braking maneuvers, the spacecraft will slow down and approach the space rocks."

- In January, Hera will release two companion probes, Milani and Juventas, to analyze material composition and peer inside the asteroid. [space.com]

  > "In January, once Hera has settled in, it will release two tiny companions that have piggybacked out to Dimorphos."

In the image: An artist's conceptual illustration of the binary asteroid system with the DART spacecraft positioned between the space rocks against a starfield.

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space.com

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SpaceX's Starship Marks 10 Years as Mars Mission Plans Face Delays

A decade after Elon Musk unveiled the Mars transport concept, Starship has reached orbit, but critical propellant transfer milestones and lunar priorities push Mars flights to at least 2028.

- Ten years after Elon Musk first unveiled his plans for a giant spaceship to carry people to Mars, the modern Starship reached orbit for the first time during its 14th test flight. [space.com]

  > "reached orbit for the first time, on Starship's 14th-ever test flight."

- While some analysts view the rocket as primarily a business tool to cut Starlink deployment costs, others believe SpaceX remains fully committed to its long-term Mars objectives. [space.com]

  > "SpaceX is still very much targeting Mars"

- A crucial engineering hurdle that must be overcome before any Mars or lunar mission is the large-scale cryogenic propellant transfer between vehicles in orbit. [space.com]

  > "the most underrated gap is the large-scale cryogenic propellant transfer between vehicles once in orbit."

- Because the necessary propellant transfer demonstration has not yet occurred, the potential 2026

Mars launch window is off the table, pushing the next opportunity to late 2028.

[space.com]

> "The next such low-energy orbital transfer opportunity for the Red Planet opens in late 2028"

In the image: The image provides an artistic illustration of a futuristic Mars base with multiple landed Starships, habitats, and a large central dome, which is not described in the text.

✓ 4 of 4 claims verified against sources · 1 source

space.com

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Astronaut Frank Rubio sets his sights on the moon

Frank Rubio, who holds the American record for the longest single spaceflight, is named as one of the four Artemis III crew members.

- Frank Rubio will be part of the four-member crew for the Artemis III mission planned for 2027. [wbur.org]

  > "The Artemis III mission is planned for 2027 to prepare for future moon landings."

- Frank Rubio holds the record for the longest single spaceflight by an American.

  [wbur.org]

  > "He holds the American record for the longest single spaceflight"

- Frank Rubio spent 371 consecutive days aboard the International Space Station during 2022-2023. [wbur.org]

  > "having spent 371 consecutive days aboard the International Space Station in 2022-2023"

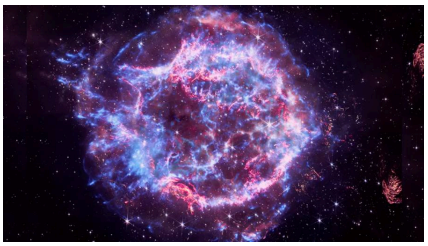
In the image: The image shows Frank Rubio waving to the crowd in a blue NASA jumpsuit during a press conference at the Johnson Space Center in Houston.

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ASTRONOMY AND ASTROPHYSICS



Some of the Universe's Biggest Explosions May Not Exist

New research suggests that stellar anomalies previously attributed to hypernovas can actually be explained by uneven, ordinary supernovas.

- Scientists discovered that hypernovas, proposed as some of the most powerful cosmic explosions, may not actually happen at all.

  [space.com]

  > “Scientists have discovered that some of the universe's most powerful explosions, hypernovas, may not happen at all.”

- Researchers found that ordinary supernovas fling out matter unevenly, which can account for stars with unusual chemical mixes.

  [space.com]

  > “The team behind these new findings suggests that ordinary supernovas could actually account for these stars when the uneven or non-spherical way they fling out matter is considered.”

- A team investigated four stars suspected of hypernova origins and concluded three were more likely created after ordinary supernovas.

  [space.com]

  > “Three of those stars were more likely to have been created after supernovas, the team concluded.”

- An asymmetrical supernova alone was found to be sufficient to explain certain odd stellar compositions without needing a hypernova. [space.com]

  > “No hypernova required. An asymmetrical supernova would be enough, the scientists found.”

In the image: The image reveals the colorful, intricate filamentary structure and expanding gas shells of a supernova remnant in deep space.

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space.com

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Amateur Astronomer Uses AI to Find Potential Super-Earth

A space science hobbyist has harnessed artificial intelligence agents to discover a potential exoplanet hiding in publicly available data from NASA's TESS telescope.

- An amateur astronomer discovered a potential hidden exoplanet using artificial intelligence agents to examine public space telescope data.

  [scientificamerican.com]

  > “An amateur astronomer harnessed artificial intelligence agents to find a previously undiscovered exoplanet lurking in publicly available data.”

- The candidate star, TIC 4206066, is located approximately 116 light-years away from Earth within the Orion constellation.

  [scientificamerican.com]

  > “The star was located around 116 light-years away from Earth in the Orion constellation”

- If confirmed, the potential exoplanet would be a super-Earth with a radius roughly 1.4 times that of our planet.

  [scientificamerican.com]

  > “that world would a super-Earth with a radius around 1.4 times as large as that of our planet.”

- Astronomer Greg Laughlin praised the discovery as a prime illustration of effective collaboration

between humans and artificial intelligence.

[scientificamerican.com]

> “Rabtsevich’s analysis is “a great example of human-AI teamwork,” he adds”

In the image: An artist's illustration of NASA's Transiting Exoplanet Survey Satellite (TESS) operating in space near a bright star and a red planet.

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scientificamerican.com

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The Biggest Stars of the 1970s and Their Careers

A look back at the most iconic actors, comedians, and television stars who defined the entertainment industry during the 1970s.

- Alan Alda became a household name when the Korean War sitcom M*A*S*H debuted in 1972.

  [mentalfloss.com]

  > “becoming a household name in the United States and beyond when Korean War sitcom M*A*S*H debuted in 1972”

- Richard Attenborough made his film directorial debut in the 1960s and was knighted in 1976. [mentalfloss.com]

  > “his directorial debut in the 1960s. By the following decade, he was one of the biggest and most acclaimed actors in show business: knighted in 1976”

- Mel Brooks won the Oscar for Best Screenplay for The Producers in 1967 before writing and directing major 1970s comedies.

  [mentalfloss.com]

  > “winning the Oscar for Best Screenplay for The Producers in 1967. The 1970s saw him write and direct some of the decade’s biggest comedies”

- David Cassidy achieved massive fame through The Partridge Family and was rumored to be the world's highest-paid entertainer.

  [mentalfloss.com]

  > “Cassidy was rumored to be the highest-paid entertainer in the world”

In the image: The image shows cast members of the television series M*A*S*H posing together in front of a camp signpost.

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mentalfloss.com

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Milky Way Planetary Systems Belong to Two Distinct Families, Study Finds

Astronomers have discovered that a star's properties dictate whether its planetary system evolves into a compact arrangement or a wide, unstable configuration.

- Researchers analyzed hundreds of multiplanet systems to uncover fundamental structural differences across the galaxy. [space.com]

  > “studied 300 of the best-understood multiplanet systems of the over 1,000 that astronomers have discovered in the Milky Way.”

- A host star's intrinsic characteristics create a direct genetic connection that shapes the evolutionary path of its planets. [space.com]

  > “Our results show that there is a genetic link between planetary systems and their stars”

- The first family consists of wide and massive planetary systems centered around host stars more massive than the Sun. [space.com]

  > “The planets in these systems orbit stars more massive than the sun with lower concentrations of elements heavier than hydrogen and helium”

- Evolutionary simulations revealed that wide, massive planetary systems tend to be dynamically unstable from birth. [space.com]

  > “The scenarios that best reproduce the dynamical properties of the family of wide and massive systems are those in which the systems are unstable from birth”

In the image: The diagram illustrates the two evolutionary pathways originating from a central star-forming nebula, showing a smaller star giving rise to a compact system with multiple small planets on the left, and a larger star evolving into a wide system with gas giants on the right.

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space.com

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PAKISTAN CRICKET TEAM



Punjab cop on anti-drone duty held with 16kg heroin near Pakistan border

timesofindia.indiatimes.com

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CRICKET

West Indies Beats India to Force T20 Series Decider

Shai Hope scored an unbeaten century to lead the West Indies to a 24-run victory in Hyderabad.

- The West Indies secured a 24-run victory against India in Hyderabad. [home.nzcity.co.nz]

  > “Shai Hope's unbeaten 102 off 50 deliveries has helped the tourists to a 24-run win in Hyderabad.”

- The touring side finished their innings with a total of 247 runs for the loss of three wickets. [home.nzcity.co.nz]

  > “The West Indies posted 247 for three, and "restricted" - if that's the right word - India to 223 all out in reply.”

- The five-match series will now conclude with a final decider fixture in Bengaluru. [home.nzcity.co.nz]

  > “The series finale is in Bengaluru on Sunday.”

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home.nzcity.co.nz

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TECHNOLOGY INDUSTRY



Virginia Tech Seeks to Snap Two-Game Skid Against Georgia Tech

After suffering back-to-back close losses, the Hokies return home to face the Yellow Jackets in an ACC matchup.

- Virginia Tech enters the matchup trying to recover from consecutive tight defeats. [921theticket.com]

  > “For the second straight week, the Hokies (4-2, 1-2) lost a close game.”

- The Hokies' defense struggled with discipline during Cal's game-winning drive. [921theticket.com]

  > “the Hokies’ defense committed two 15-yard penalties, including a pass interference call, on consecutive plays”

- Georgia Tech's linebacker corps faced a major setback due to an unexpected retirement. [921theticket.com]

  > “it suffered a big blow with the sudden retirement of linebacker Kyle Efford due to repeated injuries”

In the image: The image shows on-field game action with players in Cal and Virginia Tech uniforms during a play, which is not described in the text.

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921theticket.com

Exploring: a Technology industry story outside your usual topics, to learn whether you want more



Museum of Work & Culture Announces Annual Baking Workshop

The museum will host its annual children's baking workshop with Wright's Dairy Farm & Bakery featuring éclair making and interactive games.

- The Museum of Work & Culture is hosting its annual children's baking workshop in partnership with Wright's Dairy Farm & Bakery. [valleybreeze.com]

  > "The Museum of Work & Culture will offer its annual Children's Baking Workshop with Wright's Dairy Farm & Bakery"

- The workshop includes an éclair-making demonstration, behind-the-scenes farm footage, a French language matching game, and baking bingo. [valleybreeze.com]

  > "This year's event will feature an éclair-making demonstration with a pastry chef from Wright's and special behind-the-scenes footage from the farm."

- Participants receive ingredients, a chef's hat, apron, recipe card, and pre-printed activity materials. [valleybreeze.com]

  > "Each participant will receive a chef's hat, apron, recipe card, pre-printed activity materials, and éclair ingredients."

- Tickets cost $25 for one adult and one child, with an additional $18 for a second child. [valleybreeze.com]

  > "Tickets are $25 for one child and one adult. Tickets are an additional $18 for a second child."

In the image: The image shows the logo for The Valley Breeze newspaper, which is not mentioned in the text.

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valleybreeze.com

Exploring: a Arts, culture and entertainment story outside your usual topics, to learn whether you want more

You're caught up.

That's today's paper. Your next edition arrives tomorrow morning.